

Principles of Object-Oriented Programming

1) Abstraction - It means hiding internal details & showing the required things.

2) Encapsulation - It is the process of grouping data in a single section.

3) Inheritance - It means designing an object or a class by re-using the properties of the existing class & object. It is same as specialization.

4) Polymorphism - It is a concept in which we can execute a single operation in different ways. It is same as generalization.

• Class V/s Object

Object is defined in terms of its properties & behaviour.

Operation of behaviours will affect the properties.

- In Java, there is an area inside main memory which is known as method area which contains all the methods.

- Data Hiding - Usually, data is hidden & the operations are made visible & operations or methods are performed over the data.

For hiding data, the data members will have stricter (private) access modifier.

- Property Methods - These are methods used to access & modify the properties (data members) of an object. They are commonly called getter & setter methods.

Getter (get) - retrieves/reads a property Setter (set) - modifies/updates a property

- Types of Properties

1) Readable & Writable

2) Readable Only

3) Writable Only

- Constructors - It is a method of a class which is automatically called whenever an object is created.

- ✓ Every class in Java will have some default constructor. If we define our own, then our own constructor is taken.

- ✓ Basically, constructors are made as public but they can be made private also.

- ✓ 2 types of constructors - 1) Parameterized Constructor
2) Non-Parameterized

- ✓ No return type of constructor.

- ✓ Constructors can be overloaded.

↓
Replacement for default constructors.

Spiral